# Stability of a prescribed repeatable completion of a quantum measurement

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**External-review draft.** The argument below is intended to be reconstructed independently. Its ingredients are established; the extent to which the prescribed-map estimate and record corollary follow from earlier results remains a question for review.

## Abstract

Let $E = (E_j)$ be a finite positive-operator-valued measurement (POVM), and let $P_j$ be its full spectral sectors at or above $1 - \delta$, where $0 < \delta < 1/2$. Assume that each nonzero effect has a nonzero sector and that the sector synthesis operator is injective, with least singular value $\alpha$. Its unweighted polar isometry determines orthogonal sectors $Q_j$. We prove that the prescribed completion $F_j = Q_j + RE_jR$, where $R = I - \sum_j Q_j$, admits a repeatable instrument and satisfies

$$\|\mathcal{M}_E - \mathcal{M}_F\|_\diamond \leq 2\min\{1, \sqrt{1-\alpha^2} + \sqrt{\delta}\}.$$

The proof gives an explicit labelled dilation and a residual spectral gap. A corollary corrects separated output records to an exactly idempotent channel. We identify the classical Jordan-product inequality used in the proof and derive a weighted alternative from established polar machinery. Explicit examples distinguish the target from projective approximation, establish a square-root threshold obstruction, and limit continuity of the prescribed selection. We make no claim of optimal constants or literature priority.

## 1. Question and scope

A finite POVM can admit a repeatable instrument without being a projection-valued measurement (PVM). The relevant criterion is that every nonzero effect have eigenvalue one (Busch et al. 1995). Such measurements have orthogonal unit eigenspaces and may retain an arbitrary POVM on their common orthogonal complement. We study a particular way to select a nearby measurement in this class while retaining the input space, labels, and compressed residual effects.

The construction uses the polar factor of a synthesis operator, a standard orthogonalization device (Higham 1986). The issue addressed here is quantitative control of the resulting prescribed measurement in the full diamond norm. The hypothesis concerns near-unit sectors and their simultaneous linear independence; it is not a global almost-projectivity assumption.

The distinction from earlier work is limited. De la Salle's orthogonalization proof already contains a closely related weighted polar construction (Salle 2022). Dilation continuity is standard (Kretschmann et al. 2008), and the noncommuting matrix inequality used below is a special case of the Strang–de Bruijn Jordan-product bounds (Bruijn 1980). Section 5 makes these relationships explicit. The paper presents a specialization with a complete argument, not a claim of a new general rounding principle.

## 2. Setting and main statement

Let $\mathcal{H} \neq \{0\}$ be a finite-dimensional complex Hilbert space. Write $\|\cdot\|$ for operator norm and $\|\cdot\|_1$ for trace norm. A POVM $E = (E_j)_{j=1}^m$, $m \geq 1$, consists of positive operators summing to $I$. Its measurement channel is

$$\mathcal{M}_E(\rho) = \sum_j \mathrm{Tr}(E_j\rho)|j\rangle\langle j|.$$

We use the unhalved diamond norm; all state distances below likewise use the unhalved trace norm. For a difference of channels the diamond norm equals the supremum of output trace-norm difference over density operators with an arbitrary reference system; reference-assisted pure inputs suffice (Watrous 2018, Theorem 3.51). Consequently its maximum is two. Channels are completely positive and trace preserving (CPTP); their adjoints are unital completely positive (UCP).

Fix $0 < \delta < 1/2$ and set

$$P_j = 1_{[1-\delta,1]}(E_j), \qquad \mathcal{K}_j = \mathrm{ran}\, P_j, \qquad \mathcal{K} = \bigoplus_j \mathcal{K}_j. \tag{1}$$

Assume $P_j \neq 0$ whenever $E_j \neq 0$. A zero effect has $P_j = 0$ and may retain its label. Define

$$T : \mathcal{K} \longrightarrow \mathcal{H}, \qquad T(x_1, \ldots, x_m) = \sum_j x_j, \qquad \alpha = \sigma_{\min}(T) > 0. \tag{2}$$

Thus $T$ is injective. Write $T = UH$, with $H = |T|$, and let $C_j$ be the coordinate projection of $\mathcal{K}$ onto $\mathcal{K}_j$. Put

$$Q_j = UC_jU^*, \qquad S = UU^*, \qquad R = I - S, \qquad F_j = Q_j + RE_jR. \tag{3}$$

**Theorem 1 (prescribed completion).** Under (1)–(2), the $Q_j$ are mutually orthogonal, have the ranks of $P_j$, and obey

$$\|P_j - Q_j\| \leq a, \qquad a := \sqrt{1-\alpha^2}. \tag{4}$$

The family $F$ in (3) is a POVM with full unit eigenspaces $Q_j$, and

$$\begin{gathered} F_iQ_j = \delta_{ij}Q_j, \\ \gamma_F := \min_j \left[1 - \|(I-Q_j)F_j(I-Q_j)\|\right] \geq \delta. \end{gathered} \tag{5}$$

It admits a repeatable instrument and satisfies

$$\boxed{\|\mathcal{M}_E - \mathcal{M}_F\|_\diamond \leq 2\min\{1, a + \sqrt{\delta}\}.} \tag{6}$$

More precisely, writing $d_0 = \sqrt{1-\delta}$ and $M = \|H\|$, the sufficient condition

$$(M - \alpha)(1 - d_0) \leq 4\alpha d_0 \tag{7}$$

implies

$$\|\mathcal{M}_E - \mathcal{M}_F\|_\diamond \leq 2\sqrt{1 - \alpha^2(1-\delta)}. \tag{8}$$

The constants contain no explicit dimension, outcome-count, or sector-rank factor. The domain remains restrictive: $\sum_j \mathrm{rank}\, P_j \leq \dim \mathcal{H}$, so there are at most $\dim \mathcal{H}$ nonzero outcomes. The estimate does not apply to generic overcomplete POVMs.

## 3. Proof of Theorem 1

### 3.1. Geometry and the residual gap

The polar factor $U$ is an isometry with $\operatorname{ran} U = \operatorname{ran} T = \sum_j \operatorname{ran} P_j$. Thus $P_j R = 0$, and the $Q_j$ are orthogonal and rank preserving. A unit vector supported in one nonzero coordinate sector has $\|Tx\| = 1$, proving $\alpha \le 1$. The spectral inequality $E_j \ge (1-\delta)P_j$ gives

$$(1-\delta)TT^* = (1-\delta)\sum_j P_j \le I, \qquad M \le d_0^{-1}. \tag{9}$$

For a nonzero sector let $\iota_j : \mathcal{K}_j \to \mathcal{K}$ be the coordinate inclusion, $T_j = T\iota_j$, and $U_j = U\iota_j$. Both $T_j$ and $U_j$ are isometries, and

$$T_j^* U_j = \iota_j^* H \iota_j \ge \alpha I_{\mathcal{K}_j}.$$

Hence their overlap has smallest singular value at least $\alpha$. It follows that $\|(I-P_j)Q_j\|$ and $\|(I-Q_j)P_j\|$ are at most $a$. For two orthogonal projections their difference norm is the maximum of these two quantities, proving (4). One may verify this identity by squaring the difference and taking its two diagonal blocks relative to the first projection. Zero sectors are immediate.

Each $F_j$ is positive, and

$$\sum_j F_j = S + R\Big(\sum_j E_j\Big)R = I.$$

Equation (5)'s sector action follows from orthogonality. Fullness of the spectral sector gives

$$E_j \le P_j + (1-\delta)(I-P_j), \qquad RE_jR \le (1-\delta)R. \tag{10}$$

On $S$, the effect $F_j$ is $Q_j$; on $R$, its norm is at most $1-\delta$. Thus $Q_j$ is its full unit eigenspace and (5) holds, including zero effects. Choosing a density operator $\tau_j$ supported in each nonzero $Q_j$ gives the instrument $\mathcal{J}_j(\rho) = \operatorname{Tr}(F_j\rho)\tau_j$. For a zero effect set $\mathcal{J}_j = 0$. Since $\operatorname{Tr}(F_i\tau_j) = \delta_{ij}$, a repetition returns the same outcome.

### 3.2. An explicit labelled dilation

On $\mathcal{Y} = \bigoplus_j \mathcal{H}$, let $\Lambda_j$ project onto label $j$, and define

$$\begin{aligned} W &: \mathcal{H} \to \mathcal{Y}, \quad Wz = (\sqrt{E_j}z)_j, \\ J &: \mathcal{K} \to \mathcal{Y}, \quad J(x_j)_j = (x_j)_j. \end{aligned} \tag{11}$$

In particular, $J$ embeds the original sector vectors. Both $W$ and $J$ are isometries. Spectral commutation allows the block operator

$$D = \bigoplus_j \sqrt{E_j}\big|_{\mathcal{K}_j}, \qquad d_0 I \le D \le I.$$

Directly from their domains,

$$W^*J = TD = UHD, \qquad J^*\Lambda_j W R = 0 \quad \text{for every } j. \tag{12}$$

Indeed, the only possible nonzero coordinate of the second expression is $P_j\sqrt{E_j}R = \sqrt{E_j}P_jR = 0$.

Set

$$V = JU^* + WR. \tag{13}$$

The cross terms in $V^*V$ vanish, so $V^*V = S + R = I$. The labelwise identities in (12), together with $J^*\Lambda_j J = C_j$, give

$$V^*\Lambda_j V = UC_jU^* + RE_jR = F_j, \qquad W^*V = UHDU^* + R. \tag{14}$$

These identities concern the particular completion (3).

### 3.3. The overlap estimate

**Lemma 2 (classical Jordan-product specialization).** If $\alpha I \le H \le MI$, $d_0 I \le D \le I$, and (7) holds, then $\mathrm{Re}(HD) \ge \alpha d_0 I$.

*Proof.* Set $X = H - \alpha I$, $Y = D - d_0 I$, $h = M - \alpha$, and $r = 1 - d_0$. For $s > 0$, positivity of $(\sqrt{s}X + s^{-1/2}Y)^2$ implies

$$XY + YX \ge -sX^2 - s^{-1}Y^2 \ge -shX - (r/s)Y.$$

Therefore

$$\mathrm{Re}(HD) - \alpha d_0 I \ge (d_0 - sh/2)X + (\alpha - r/(2s))Y.$$

For $h > 0$, take $s = 2d_0/h$; both coefficients are nonnegative by (7). For $h = 0$, $H = \alpha I$ and the assertion is immediate. □

For attribution, put $u = (M - \alpha)/(M + \alpha)$ and $v = (1 - d_0)/(1 + d_0)$. Condition (7) is exactly $u + v \le 1$; the corresponding branch of de Bruijn's Theorem 3.2 and Sections 4–5 gives the lower bound $(M + \alpha)(1 + d_0)(1 - u)(1 - v)/4 = \alpha d_0$ (Bruijn 1980). The proof above is included for completeness.

By (14) and $\alpha d_0 \le 1$, Lemma 2 yields

$$\mathrm{Re}(W^* V) \ge \alpha d_0 S + R \ge \alpha d_0 I.$$

For any normalized $\psi \in \mathcal{H} \otimes \mathcal{A}$, the dilated vectors $(W \otimes I_{\mathcal{A}})\psi$ and $(V \otimes I_{\mathcal{A}})\psi$ have overlap of modulus at least $\alpha d_0$. The trace distance of their pure density operators is consequently at most $2\sqrt{1 - \alpha^2 d_0^2}$. Apply to both the same CPTP label readout $\mathcal{L}(X) = \sum_j \mathrm{Tr}(\Lambda_j X)|j\rangle\langle j|$, with the identity on $\mathcal{A}$. Trace-norm contractivity and the reference-assisted characterization prove (8).

To obtain (6), write $b = \sqrt{\delta}$. If $a + b \ge 1$, use the universal bound two. Otherwise $a < 1 - b$, so $\alpha^2 > 2b - b^2 > b^2$, hence $\alpha > b$. By (9),

$$(M - \alpha)(1 - d_0) \le M(1 - d_0) \le \frac{\delta}{d_0(1 + d_0)} \le \delta < 4\alpha d_0.$$

Here $d_0 > 1/\sqrt{2}$, and $b < 4d_0$. Thus (7) holds, and

$$2\sqrt{1 - \alpha^2(1 - \delta)} = 2\sqrt{a^2 + \alpha^2 \delta} \le 2(a + b).$$

This completes the proof of Theorem 1. □

## 4. Correction of supplied records

Let $\sigma_j$ be density operators on the same $\mathcal{H}$, and set

$$\Phi(\rho) = \sum_j \mathrm{Tr}(E_j \rho)\sigma_j, \qquad \eta = \|\Phi^2 - \Phi\|_\diamond.$$

For this section discard zero effects and their unused records. Assume at least two remaining labels and define

$$\kappa = \min_j \frac{1}{2} \inf_{\omega \in \mathrm{conv}\{\sigma_i : i \ne j\}} \|\sigma_j - \omega\|_1 > 0, \qquad e = \max_j (1 - \lambda_{\max}(E_j)). \tag{15}$$

The main hypotheses give $0 \le e \le \delta < 1/2$. With one remaining label the channel is constant and already idempotent.

**Corollary 3 (record correction).** Under Theorem 1 and (15), there are explicit states $\tau_j$ supported in $Q_j$ such that $\Pi(\rho) = \sum_j \operatorname{Tr}(F_j\rho)\tau_j$ is exactly idempotent. For any valid measurement movement bound $B_{\mathrm{move}}$,

$$\boxed{\|\Phi - \Pi\|_\diamond \le \min\left\{2,\ B_{\mathrm{move}} + 2a + \sqrt{\frac{2\eta}{\kappa\delta(1-2e)}}\right\}.} \tag{16}$$

In particular,

$$\|\Phi - \Pi\|_\diamond \le \min\left\{2,\ 4a + 2\sqrt{\delta} + \sqrt{\frac{2\eta}{\kappa\delta(1-2\delta)}}\right\}. \tag{17}$$

*Proof.* Put $v_j = \Phi(\sigma_j) - \sigma_j$ and $D_{\mathrm{rec}} = \max_j \|v_j\|_1$. The action of $\Phi^2 - \Phi$ on a reference-assisted state is $\sum_j v_j \otimes B_j$, where the conditional operators $B_j$ are positive and $\sum_j \operatorname{Tr} B_j = 1$. Thus $\eta \le D_{\mathrm{rec}}$. Choose a maximizing label and an input whose probability of that label is at least $1 - e$. The reverse triangle inequality gives

$$(1-2e)D_{\mathrm{rec}} \le \eta \le D_{\mathrm{rec}}. \tag{18}$$

Writing $a_j = \operatorname{Tr}(E_j\sigma_j)$, when $a_j < 1$ the other outcomes yield a state $\omega_j$ in the competing convex hull with $v_j = (1-a_j)(\omega_j - \sigma_j)$. The case $a_j = 1$ is immediate. Hence

$$2\kappa(1-a_j) \le \|v_j\|_1, \qquad 1 - \operatorname{Tr}(P_j\sigma_j) \le q := \frac{\eta}{2\kappa\delta(1-2e)}, \tag{19}$$

where the second inequality uses (10).

For a purification of $\sigma_j$, the triangle inequality and (4) imply

$$\sqrt{1 - \operatorname{Tr}(Q_j\sigma_j)} \le \sqrt{1 - \operatorname{Tr}(P_j\sigma_j)} + \|P_j - Q_j\| \le \sqrt{q} + a.$$

If $z_j = \operatorname{Tr}(Q_j\sigma_j) > 0$, set $\tau_j = Q_j\sigma_jQ_j/z_j$. Projecting the purification gives a normalized vector with overlap $\sqrt{z_j}$ with the original; the pure-state trace distance formula and partial trace therefore give

$$\|\sigma_j - \tau_j\|_1 \le 2\sqrt{1 - z_j} \le 2\sqrt{q} + 2a. \tag{20}$$

This is the standard normalized gentle-projection estimate (Watrous 2018, Corollary 3.15 and Theorem 3.33). If $z_j = 0$, the preceding inequality forces $\sqrt{q} + a \ge 1$; choose $\tau_j = Q_j/\operatorname{rank} Q_j$, which satisfies (20).

Now $\operatorname{Tr}(F_i\tau_j) = \delta_{ij}$, so $\Pi^2 = \Pi$. Keeping the records $\sigma_j$ while changing $E$ to $F$ costs at most $B_{\mathrm{move}}$, by common CPTP post-processing. Keeping $F$ while changing its records costs at most $\max_j \|\sigma_j - \tau_j\|_1$, also for inputs with a reference. Combining these facts proves (16). Theorem 1 and $e \le \delta$ give (17). □

For a concrete nontrivial regime, take $0 < \varepsilon < 1/16$ and $E_0 = \operatorname{diag}(1-\varepsilon, \varepsilon)$, $E_1 = I - E_0$, and the two computational-basis records. Then $\alpha = \kappa = 1$, $e = \varepsilon$, $\eta = 2\varepsilon(1-2\varepsilon)$, and the constructed channel is complete dephasing, at distance $2\varepsilon$. Choosing $\delta = \sqrt{\varepsilon}$ in (16) with $B_{\mathrm{move}} = 2\sqrt{\delta}$ gives $4\varepsilon^{1/4}$, which is below two for $\varepsilon < 1/16$. The certificate is useful but can be very conservative.

## 5. Relationship to existing constructions

**Proposition 4 (weighted comparison).** Keep the hypotheses and notation of Theorem 1, but let $TD = \widehat{U}|TD|$. Then, without condition (7),

$$\begin{aligned} \widehat{F}_j &= \widehat{U}C_j\widehat{U}^* + RE_jR, \\ \|\mathcal{M}_E - \mathcal{M}_{\widehat{F}}\|_\diamond &\le 2\sqrt{1 - \alpha^2(1-\delta)}. \end{aligned} \tag{21}$$

*Proof.* The invertibility of $D$ gives $\operatorname{ran}(TD) = \operatorname{ran} T$ and $|TD| \geq \alpha d_0 I$. Thus $\widehat{U}$ has the same range projection $S$. Equations (12)–(14), with $\widehat{V} = J\widehat{U}^* + WR$, give the isometry and effect identities, while

$$W^*\widehat{V} = \widehat{U}|TD|\widehat{U}^* + R \geq \alpha d_0 I.$$

The same reference-assisted overlap argument proves (21). □

This is a reconstructed application of established polar alignment, not a claim that a cited paper states (21) verbatim. Its connection to de la Salle's proof is explicit. On $S\mathcal{H}$, put $A_j = SE_jS$. The $P_j$ commute with their $A_j$, have total rank $\dim S\mathcal{H}$, and satisfy

$$\sum_j P_j A_j \geq (1-\delta)\alpha^2 I_{S\mathcal{H}}.$$

They can therefore replace the state-selected projections in the finite proof's polar step. That column becomes $x = (P_j\sqrt{A_j})_j = DT^*$; its adjoint is $TD$ (Salle 2022, Lemma 3.1 and Section 3). The residual extension and overlap calculation above make the application uniform. Broad existence of a nearby same-label repeatable measurement consequently has a short derivation from this machinery. The weighted construction generally differs from (3); (21) does not prove Theorem 1's estimates for that prescribed map.

KSW's continuity theorem supplies the generic dilation-to-channel comparison (Kretschmann et al. 2008, Proposition 3). Kotowski–Oszmaniec's dimension-deficient Naimark construction proves projective simulability for a different complement-noise completion, without requiring unit eigenspaces or asserting the present diamond estimate (Kotowski and Oszmaniec 2025, Theorem 6).

The exact record endpoint also has established structure. On the algebra of active labels define $\Delta(f) = \sum_j f_j F_j$ and $\Upsilon(X) = (\operatorname{Tr}(\tau_j X))_j$. They are UCP maps with $\Upsilon\Delta = I$ and $\Pi^* = \Delta\Upsilon$, a commutative instance of exact encoding/decoding structure (Kitaev 2025, Theorem 1.2). Kitaev's Theorem 12.3 gives general approximate factors; it does not directly supply this particular exact correction of the original effects and records. The additional specification here is quantitative and applies to a substantially narrower class.

These comparisons leave the prescribed-map estimate and record correction as the appropriate subjects for an external implication and priority review. They do not establish that either specialization is new.

# 6. Counterexamples and limitations

## 6.1. Repeatable approximation differs from projective approximation

On $\mathbb{C}^3$, let

$$E_0(t) = \operatorname{diag}(1-t, t, 1/2), \quad E_1(t) = I - E_0(t), \qquad 0 < t < 1/2.$$

For $\delta = t$, the sectors are the first two coordinate lines and $\alpha = 1$. The prescribed completion is $F_0 = \operatorname{diag}(1, 0, 1/2)$, $F_1 = I - F_0$, with distance $2t$. However, its input measurement has distance exactly one to the nearest binary PVM. For any projection $L$, ordered-eigenvalue perturbation gives $\|E_0(t) - L\| \geq 1/2$, because the middle eigenvalue of $E_0(t)$ is $1/2$ whereas that of $L$ is zero or one. Binary measurement-channel distance is $2\|E_0(t) - L\|$; the upper bound one is attained by $L = \operatorname{diag}(1, 0, 0)$. The binary norm identity follows from the norm of the Hermitian functional $\rho \mapsto \operatorname{Tr}((E_0 - L)\rho)$; an extremal eigenvector attains it, and reference systems do not increase it. Also $\|I - \sum_j E_j(t)^2\| = 1/2$.

Thus small $\delta$ and perfect packing do not imply global near-projectivity. The example does not preclude the packed-subspace reduction in Section 5.

### 6.2. A square-root threshold obstruction

**Proposition 5.** For every $0 < \delta < 1/10$, there is a three-outcome POVM on $\mathbb{C}^4$ satisfying Theorem 1 with $\alpha = 1$, such that every repeatable POVM $G$ on the same space and with the same three labels obeys

$$\|\mathcal{M}_E - \mathcal{M}_G\|_\diamond \geq \frac{\sqrt{\delta}}{4\sqrt{3}}. \tag{22}$$

*Proof.* In the basis $e_1, \dots, e_4$, put $t = \sqrt{\delta}/4$ and

$$E_1 = \begin{pmatrix} 1-\delta/2 & 0 & 0 & 0 \\ 0 & \delta/2 & 0 & t \\ 0 & 0 & 0 & 0 \\ 0 & t & 0 & 1/4 \end{pmatrix}, \tag{23}$$
$$E_2 = \operatorname{diag}(\delta/2, 1-\delta, \delta/2, 1/2), \quad E_3 = I - E_1 - E_2.$$

The nontrivial low $2 \times 2$ blocks of $E_1, E_3$ have determinant $\delta/16 > 0$ and trace $1/4 + \delta/2 < 3/10$. The remaining entries are nonnegative. Thus these are POVM effects, and their full near-unit sectors are respectively $\mathbb{C}e_1, \mathbb{C}e_2, \mathbb{C}e_3$; hence $\alpha = 1$.

Set $\rho_0 = \max_j \|E_j - G_j\|$. If $\rho_0 \leq 1/4$, then $G_2 \neq 0$, since $\|E_2\| = 1 - \delta > 9/10$. Repeatability supplies a unit vector $v$ with $G_2 v = v$, and positivity of the other effects gives $G_1 v = 0$. Write $v = a_2 e_2 + b_2$, $b_2 \perp e_2$. Since $I - E_2 \geq I/2$ on $e_2^\perp$,

$$\|(I - E_2)v\| \leq \rho_0 \quad \Longrightarrow \quad \|b_2\| \leq 2\rho_0, \quad |a_2| \geq \sqrt{3}/2.$$

Moreover, $\|E_1 v\| \leq \rho_0$, $\|E_1\| \leq 1$, and $\|E_1 e_2\| \geq t$. Therefore

$$|a_2| t \leq \|a_2 E_1 e_2\| \leq \rho_0 + \|b_2\| \leq 3\rho_0,$$

so $\rho_0 \geq t/(2\sqrt{3})$. For any two POVMs, a state maximizing one effect difference gives $\|\mathcal{M}_E - \mathcal{M}_G\|_\diamond \geq 2\rho_0$: the other probability differences sum to its negative. This proves (22). If $\rho_0 > 1/4$, the same conclusion follows from distance greater than $1/2$. □

The proposition establishes the necessity of square-root threshold order for this same-space, same-label target class. It does not establish optimality of the constant two, the dependence on $\alpha$, or the joint bound.

### 6.3. Other limits

The strict accessibility condition in (18) is essential. For $E_0 = E_1 = I/2$ and orthogonal pure records, $e = 1/2$, $\Phi$ is constant, $\eta = 0$, but $D_{\mathrm{rec}} = 1$. This example violates the main theorem's near-unit hypothesis; it concerns the accessibility lemma's endpoint.

The prescribed spectral selection is not globally continuous. Fix $\delta = 1/5$ and

$$E_0(\varepsilon) = \operatorname{diag}(1, 4/5 - \varepsilon, 0, 1/2), \qquad E_1(\varepsilon) = I - E_0(\varepsilon).$$

At $\varepsilon = 0$, $F_0 = \operatorname{diag}(1, 1, 0, 1/2)$. For $0 < \varepsilon < 1/10$, $F_0(\varepsilon) = E_0(\varepsilon)$. All examples satisfy the hypotheses with $\alpha = 1$. Input measurement distance $2\varepsilon$ tends to zero, while the completed measurement distance tends to $2/5$. A perturbation-continuity claim therefore requires additional threshold isolation.

Finally, (6) is informative only when $a + \sqrt{\delta} < 1$. The record certificate can become trivial for small $\kappa$ or unfavorable $\eta/\delta$. No experimental estimation procedure for all these parameters, nor practical advantage of the unweighted over the weighted completion, is established here.

## 7. Reproducibility, methodology, and review status

The RC-Discovery repository contains the frozen statement, analytic source notes, numerical identity checks, reference-assisted primal/dual SDPs, negative controls, and recorded seeds and source hashes. Its theorem-level comparison was checked on 104 weighted constructions and six primal/dual SDPs, alongside exact rational examples. These computations support finite instances; the proofs above do not rely on solver output. The Lean development checks 13 constituent declarations, not the full spectral and channel theorem.

This work was developed using **VIBE MATH**, an AI-assisted workflow of conjecture generation, derivation, adversarial checking, literature comparison, and computational support directed by Christopher Gardner. AI systems are research tools, not authors or endorsers. Agreement among internal AI-assisted checks is not external verification.

The requested external assessment is concrete: reconstruct Theorem 1 and Corollary 3 from their stated hypotheses, check the scope of the counterexamples, and determine whether the prescribed-map estimate and record correction are already stated or follow as short corollaries of existing results. Correctness, priority, optimality, and usefulness are separate questions. This draft has not undergone independent specialist review and is not represented as submission-ready.

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